

Grounding and Bonding — Two Different Jobs

Three near-identical phrases that the exam relies on you mixing up, plus the two conductor sizes it asks for by number.

Grounding connects the system to earth. Bonding ties metal parts together so fault current has a continuous path back to the source. They sound alike and they do different work, and every question in this lesson turns on which one is being described.

YOU HAVE MISSED THIS 3 TIMES

The grounding electrode system:

Provides an alternate return path for electricity.

That is the definition to hold: an alternate return path, a low-resistance route for fault current to earth. It is not primarily there to protect people from shock — that is the GFCI's job, and the exam offers the shock-protection wording precisely because it sounds responsible.

Bonding	What ensures continuity of the fault path, carries fault current safely, and helps the overcurrent device operate. If a question lists all three of those clauses, the answer is bonding — earthing alone does not trip breakers.
Confirming a bond	With an ohmmeter. Bonding is continuity, continuity is low resistance, so match the instrument to the quantity being measured.
Ungrounded conductor	The formal name for the hot, or live, wire.
Grounded conductor	The neutral.
Grounding conductor	The safety ground.

Three phrases, one word apart. Ungrounded means hot. Grounded means neutral. Grounding means ground.

The two sizes asked for by number

The numbers here

Grounding electrode conductor, 200-amp service with a UFER (concrete-encased) electrode

4 AWG copper

Pool equipotential bonding conductor at the pump motor terminal

8 AWG SOLID copper

YOU HAVE MISSED THIS 2 TIMES

The grounding electrode conductor requires a #_____ AWG copper wire for UFER grounding on a 200-amp service.

Four.

Where the two systems must stay apart

In a subpanel	Neutrals and grounds must be separated onto isolated bars.
At the service equipment	They are bonded together once, and only once, by the main bonding jumper.
A detached building	Fed by a feeder, it needs its own grounding electrode.

YOU HAVE MISSED THIS 2 TIMES

A subpanel for a detached building has melted insulation on a neutral conductor. What is the most likely cause?

The neutral conductor is loose at the terminal bar.

This is the heat rule from lesson 1 applied to a real finding. Loose means high resistance, high resistance means heat, and heat melts the insulation nearest the bad connection.

Say it back

Cover the page. Answer each one out loud.

1. Define the grounding electrode system in one phrase.
2. Which does the fault-path-continuity description belong to, grounding or bonding?
3. Ungrounded, grounded, grounding — which is hot, neutral, ground?
4. GEC size for a 200-amp UFER service?
5. Pool bonding conductor size, and which word must not be dropped?
6. In a subpanel, what happens to neutrals and grounds, and why?
7. Melted insulation on a subpanel neutral. Most likely cause?